

▼ 타이타닉 데이터 분석 및 시각화

```
import numpy as np
import pandas as pd
import seaborn as sns
```

```
sns.set_style('whitegrid')
```

```
titanic = sns.load_dataset('titanic')
```

```
titanic.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult
0	0	3	male	22.0	1	0	7.2500	S	Third	man	0
1	1	1	female	38.0	1	0	71.2833	C	First	woman	0
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	0
3	1	1	female	35.0	1	0	53.1000	S	First	woman	0
4	0	3	male	35.0	0	0	8.0500	S	Third	man	0

```
titanic.describe()
```

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
titanic.dropna().describe()
```

	survived	pclass	age	sibsp	parch	fare
count	182.000000	182.000000	182.000000	182.000000	182.000000	182.000000
mean	0.675824	1.192308	35.623187	0.467033	0.478022	78.919735
std	0.469357	0.516411	15.671615	0.645007	0.755869	76.490774
min	0.000000	1.000000	0.920000	0.000000	0.000000	0.000000
25%	0.000000	1.000000	24.000000	0.000000	0.000000	29.700000
50%	1.000000	1.000000	28.000000	0.000000	0.000000	53.000000

```
titanic.var()
```

```
survived      0.236772
pclass        0.699015
age           211.019125
sibsp         1.216043
parch         0.649728
fare          2469.436846
adult_male    0.239723
alone         0.239723
dtype: float64
```

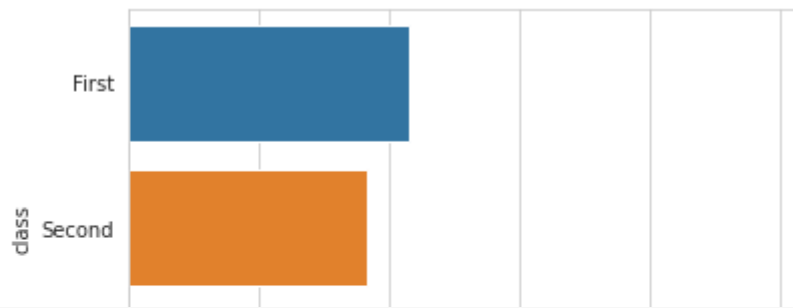
```
titanic.mad()
```

```
survived      0.473013
pclass        0.761968
age           11.322944
sibsp         0.713780
parch         0.580742
fare          28.163692
adult_male    0.478908
alone         0.478908
dtype: float64
```

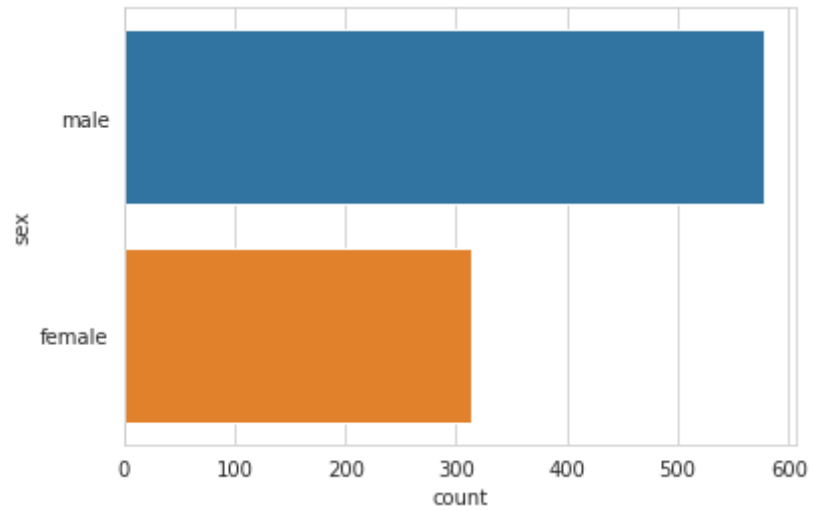
```
titanic.groupby('class').count()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	who	adult_male
class										
First	216	216	216	186	216	216	216	214	216	216
Second	184	184	184	173	184	184	184	184	184	184
Third	491	491	491	355	491	491	491	491	491	491

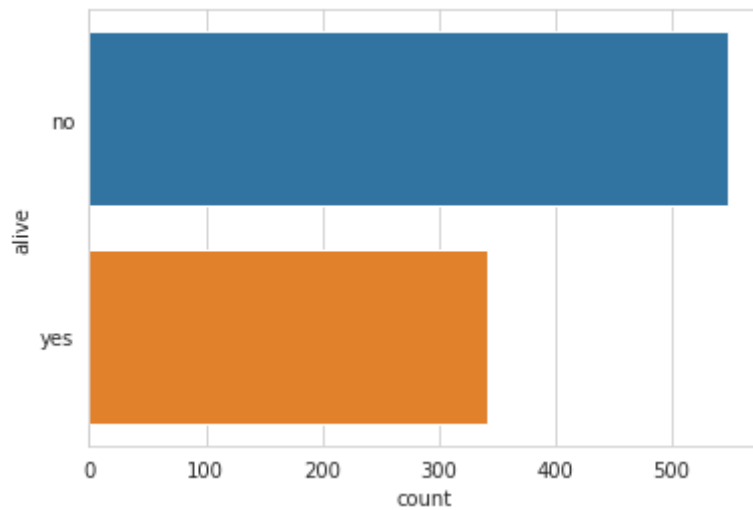
```
sns.countplot(y='class', data=titanic);
```



```
sns.countplot(y='sex', data=titanic);
```



```
sns.countplot(y='alive', data=titanic);
```



```
sns.countplot(y='alone', data=titanic);
```



```
titanic.groupby('class').std()
```

	survived	pclass	age	sibsp	parch	fare	adult_male
class							
First	0.484026	0.0	14.802856	0.611898	0.693997	78.380373	0.498555
Second	0.500623	0.0	14.001077	0.601633	0.690963	13.417399	0.499911
Third	0.428949	0.0	12.495398	1.374883	0.888861	11.778142	0.477552

```
titanic.groupby('class')['fare'].median()
```

```
class
First    60.2875
Second   14.2500
Third     8.0500
Name: fare, dtype: float64
```

```
titanic.query("alive == 'yes'")
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who
1	1	1	female	38.0	1	0	71.2833	C	First	woman
2	1	3	female	26.0	0	0	7.9250	S	Third	woman
3	1	1	female	35.0	1	0	53.1000	S	First	woman
8	1	3	female	27.0	0	2	11.1333	S	Third	woman
9	1	2	female	14.0	1	0	30.0708	C	Second	child
...	...	...	...	...	...	...	...	...	...	...
875	1	3	female	15.0	0	0	7.2250	C	Third	child
879	1	1	female	56.0	0	1	83.1583	C	First	woman
880	1	2	female	25.0	0	1	26.0000	S	Second	woman
887	1	1	female	19.0	0	0	30.0000	S	First	woman
889	1	1	male	26.0	0	0	30.0000	C	First	man

342 rows × 15 columns

```
titanic.query("alive == 'yes'").groupby('class').count()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	who	adult_male
class										
First	136	136	136	122	136	136	136	134	136	136
Second	87	87	87	83	87	87	87	87	87	87

```
titanic.groupby('class')['age'].describe()
```

	count	mean	std	min	25%	50%	75%	max
class								
First	186.0	38.233441	14.802856	0.92	27.0	37.0	49.0	80.0
Second	173.0	29.877630	14.001077	0.67	23.0	29.0	36.0	70.0
Third	355.0	25.140620	12.495398	0.42	18.0	24.0	32.0	74.0

```
titanic.query("alive == 'yes'").groupby('class').describe()
```

	survived								pclass					
	count	mean	std	min	25%	50%	75%	max	count	mean	std	min	25%	50%
class														
First	136.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	136.0	1.0	0.0	1.0	1.0	1.0
Second	87.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	87.0	2.0	0.0	2.0	2.0	2.0
Third	119.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	119.0	3.0	0.0	3.0	3.0	3.0

```
titanic.groupby('sex')['age'].aggregate([min, np.median, max])
```

	min	median	max
sex			
female	0.75	27.0	63.0
male	0.42	29.0	80.0

```
titanic.query("age > 30").groupby('class').median()
```

	survived	pclass	age	sibsp	parch	fare	adult_male	alone
class								
First	1	1	45.0	0	0	56.9292	True	False
Second	0	2	39.0	0	0	14.0000	True	True
Third	0	3	38.0	0	0	8.0500	True	True

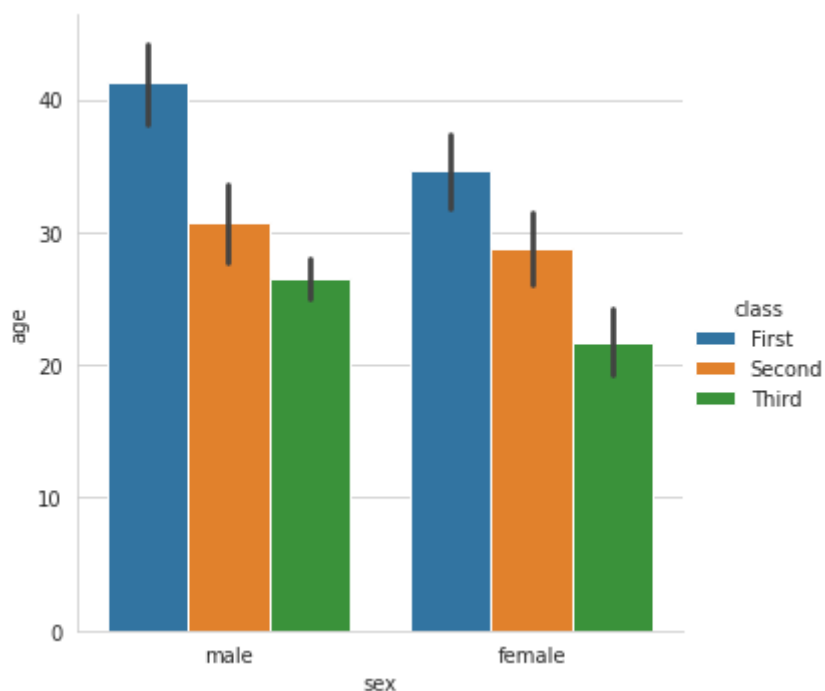
```
titanic.query("fare < 20").groupby('class').median()
```

	survived	pclass	age	sibsp	parch	fare	adult_male	alone
class								
First	0	1	38.5	0	0	0.0000	True	True
Second	0	2	30.0	0	0	13.0000	True	True
Third	0	3	25.0	0	0	7.8958	True	True

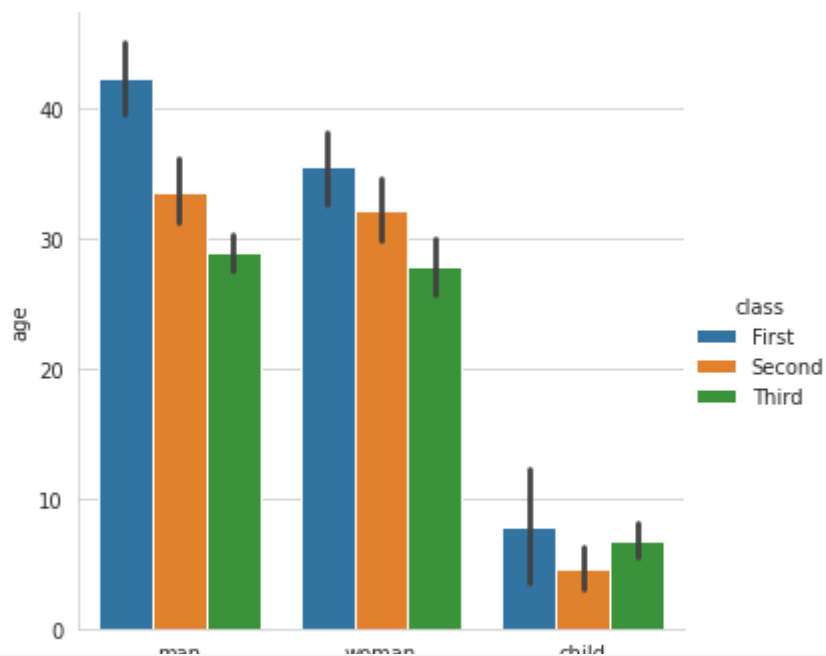
```
titanic.groupby(['class', 'sex'])['age'].mean().unstack()
```

	sex	female	male
class			
First		34.611765	41.281386
Second		28.722973	30.740707
Third		21.750000	26.507589

```
sns.catplot(x='sex', y='age',
            hue='class', kind='bar',
            data=titanic);
```



```
sns.catplot(x='who', y='age',
            hue='class', kind='bar',
            data=titanic);
```



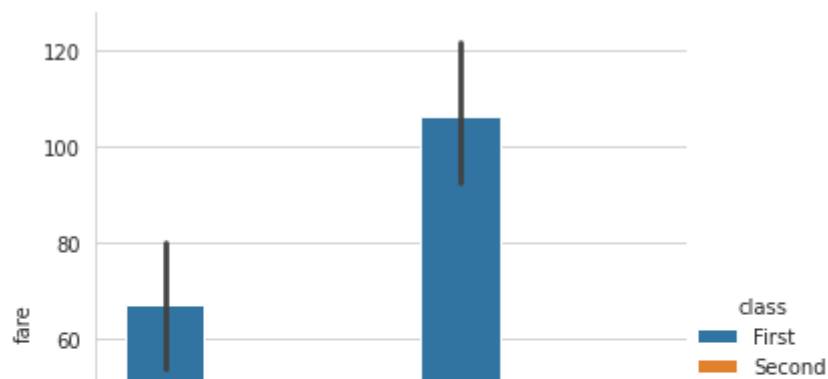
```
titanic.groupby(['class', 'sex'])['fare'].mean().unstack()
```

sex	female	male
class		
First	106.125798	67.226127
Second	21.970121	19.741782
Third	16.118810	12.661633

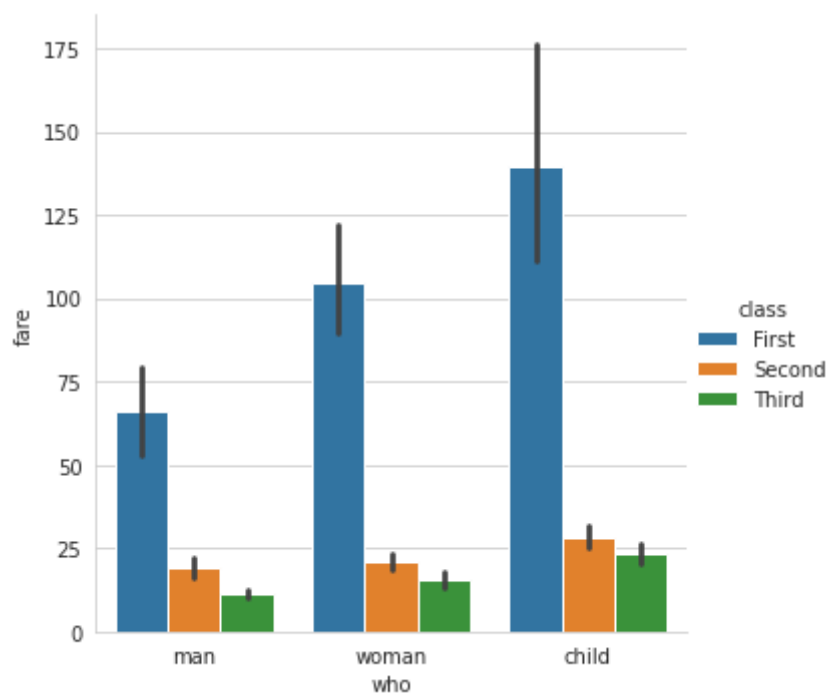
```
titanic.groupby(['class', 'who'])['fare'].mean().unstack()
```

who	child	man	woman
class			
First	139.382633	65.951086	104.317995
Second	28.323905	19.054124	20.868624
Third	23.220190	11.340213	15.354351

```
sns.catplot(x='sex', y='fare',
            hue='class', kind='bar',
            data=titanic);
```



```
sns.catplot(x='who', y='fare',
            hue='class', kind='bar',
            data=titanic);
```



```
titanic.groupby(['class', 'sex'])['survived'].mean().unstack()
```

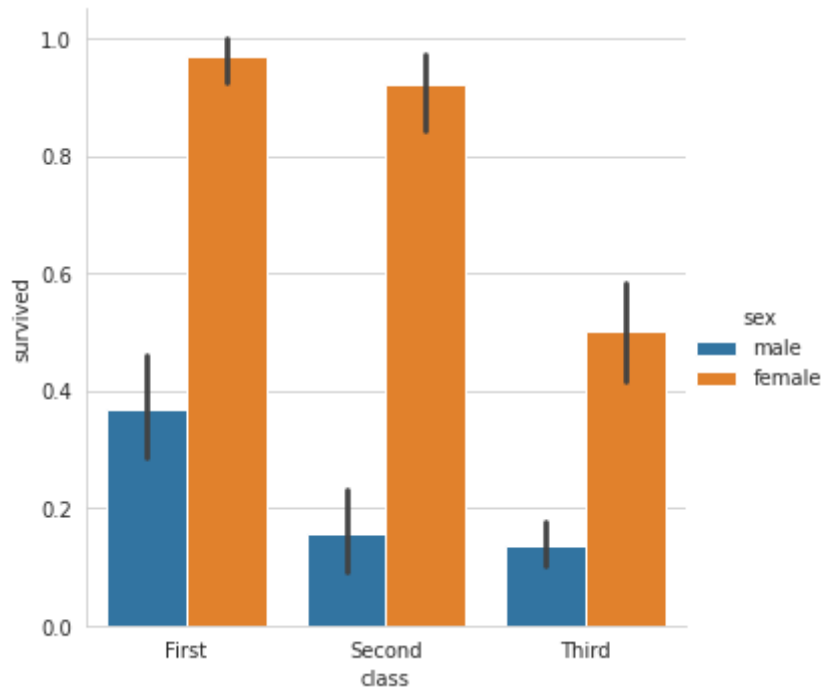
sex	female	male
class		
First	0.968085	0.368852
Second	0.921053	0.157407
Third	0.500000	0.135447

```
titanic.pivot_table('survived', index='class', columns='who')
```

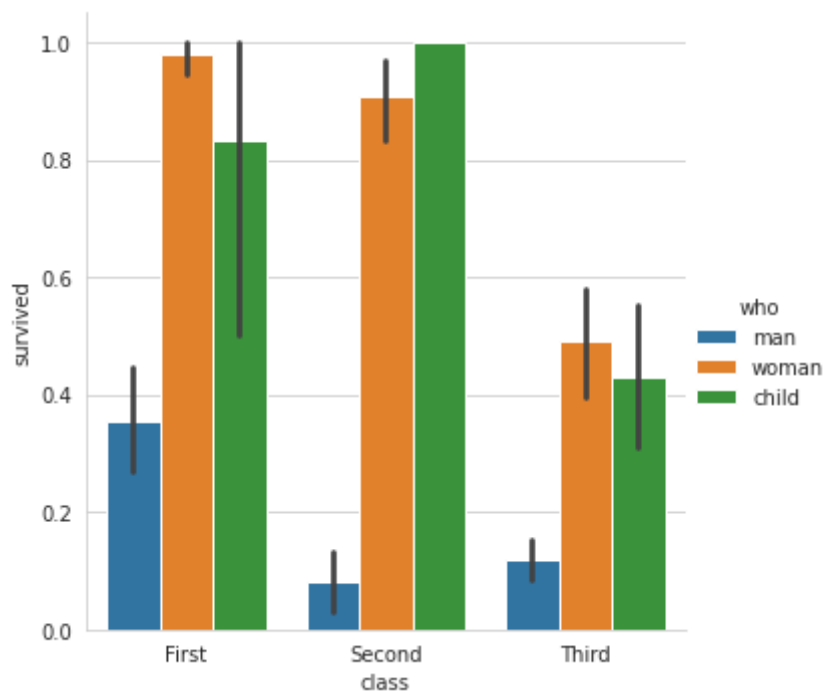
who child man woman

class

```
sns.catplot(x='class', y='survived',  
            hue='sex', kind='bar',  
            data=titanic);
```



```
sns.catplot(x='class', y='survived',  
            hue='who', kind='bar',  
            data=titanic);
```



```
age = pd.cut(titanic['age'], [0, 18, 40, 80])  
titanic.pivot_table('survived', ['who', age], 'class')
```

	class	First	Second	Third
who	age			
child	(0, 18]	0.833333	1.000000	0.431034
man	(0, 18]	0.500000	0.000000	0.086957
	(18, 40]	0.478261	0.063492	0.146199
	(40, 80]	0.280000	0.095238	0.064516
woman	(0, 18]	1.000000	1.000000	0.461538
	(18, 40]	0.979167	0.914894	0.480000
	(40, 80]	0.961538	0.846154	0.111111

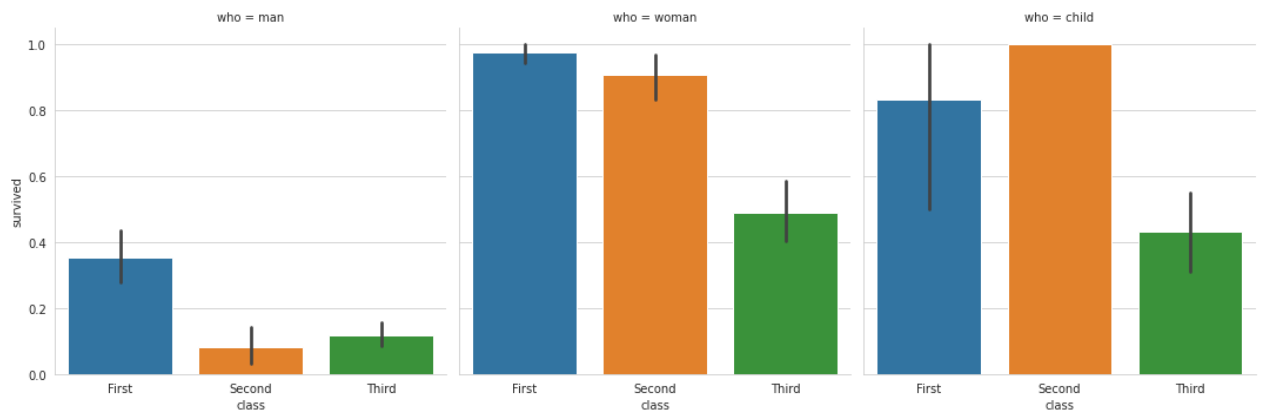
```
fare = pd.qcut(titanic['fare'], 3)
titanic.pivot_table('survived', ['who', 'age'], [fare, 'class'])
```

	fare	(-0.001, 8.662]		(8.662, 26.0]			(26.0, 512.329]		
	class	First	Third	First	Second	Third	First	Second	Third
who	age								
child	(0, 18]	NaN	0.666667	NaN	1.000000	0.703704	0.833333	1.0	0.08
man	(0, 18]	NaN	0.125000	NaN	0.000000	0.000000	0.500000	0.0	0.00
	(18, 40]	0.0	0.138462	NaN	0.078431	0.114286	0.523810	0.0	0.50
	(40, 80]	NaN	0.074074	0.0	0.111111	0.000000	0.291667	0.0	NaN
woman	(0, 18]	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0

```
titanic.pivot_table('survived', index='who', columns='class', margins=True)
```

class	First	Second	Third	All
who				
child	0.833333	1.000000	0.431034	0.590361
man	0.352941	0.080808	0.119122	0.163873
woman	0.978022	0.909091	0.491228	0.756458
All	0.629630	0.472826	0.242363	0.383838

```
sns.catplot(x='class', y='survived',
            col='who', kind='bar',
            data=titanic);
```



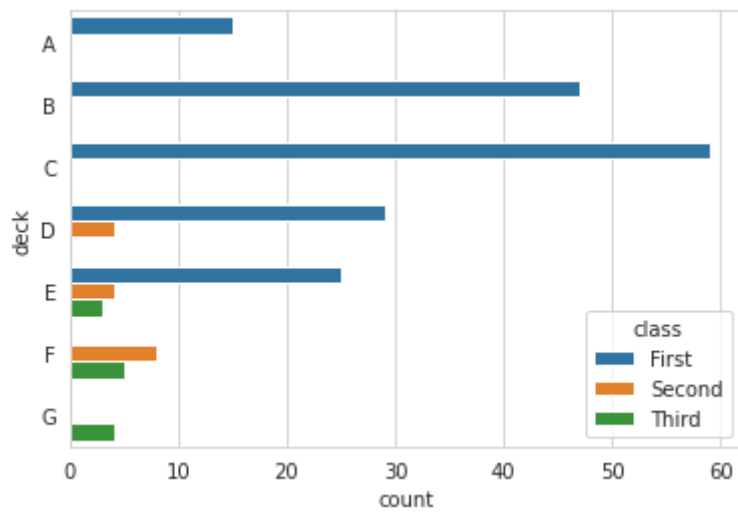
```
titanic.pivot_table('survived', index='deck', columns='class', margins=True)
```

class	First	Second	Third	All
deck				
A	0.466667	NaN	NaN	0.466667
B	0.744681	NaN	NaN	0.744681
C	0.593220	NaN	NaN	0.593220
D	0.758621	0.7500	NaN	0.757576
E	0.720000	0.7500	1.0	0.750000
F	NaN	0.8750	0.2	0.615385
G	NaN	NaN	0.5	0.500000
All	0.668571	0.8125	0.5	0.669951

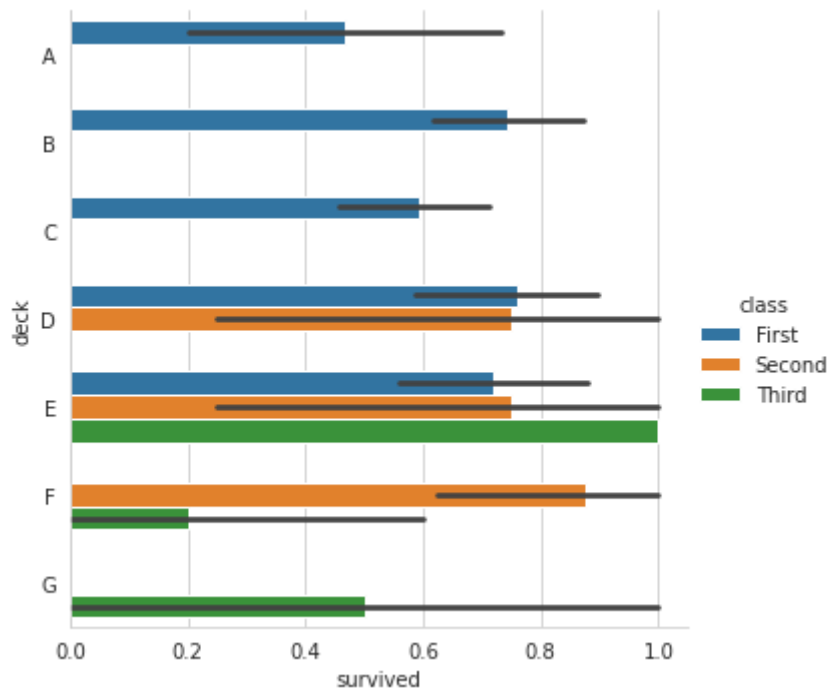
```
sns.countplot(y='deck', data=titanic);
```



```
sns.countplot(y='deck', hue='class', data=titanic);
```



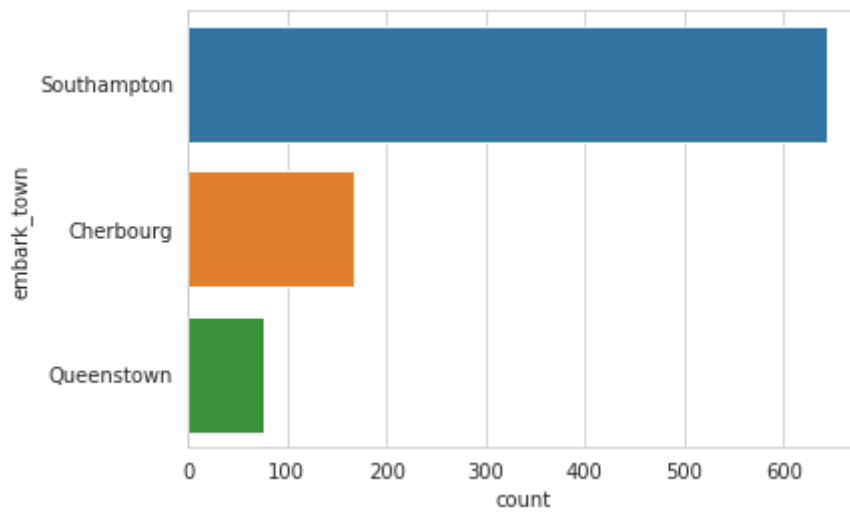
```
sns.catplot(x='survived', y='deck',  
            hue='class', kind='bar',  
            data=titanic);
```



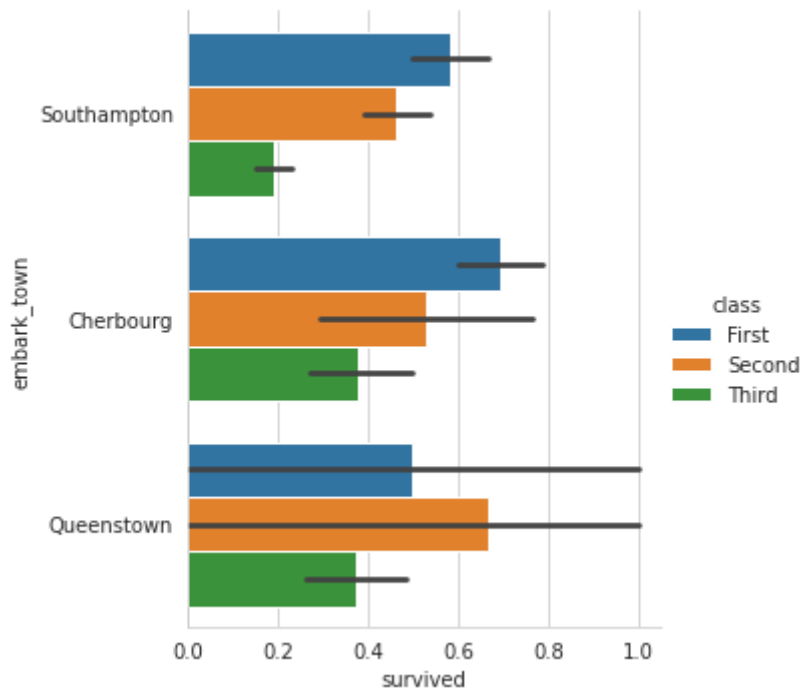
```
titanic.pivot_table('survived', index='embark_town', columns='class', margins=True)
```

class First Second Third All

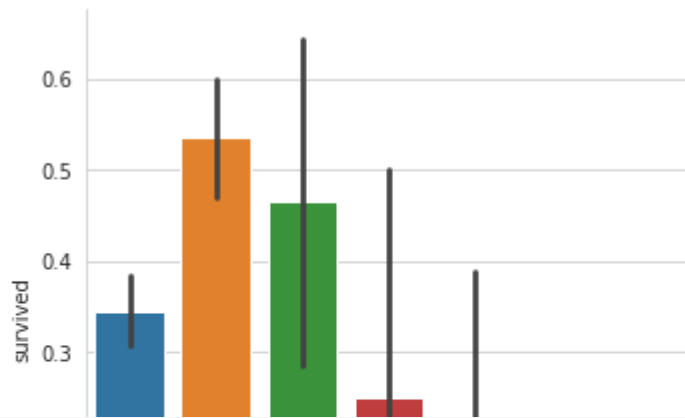
```
sns.countplot(y='embark_town', data=titanic);
```



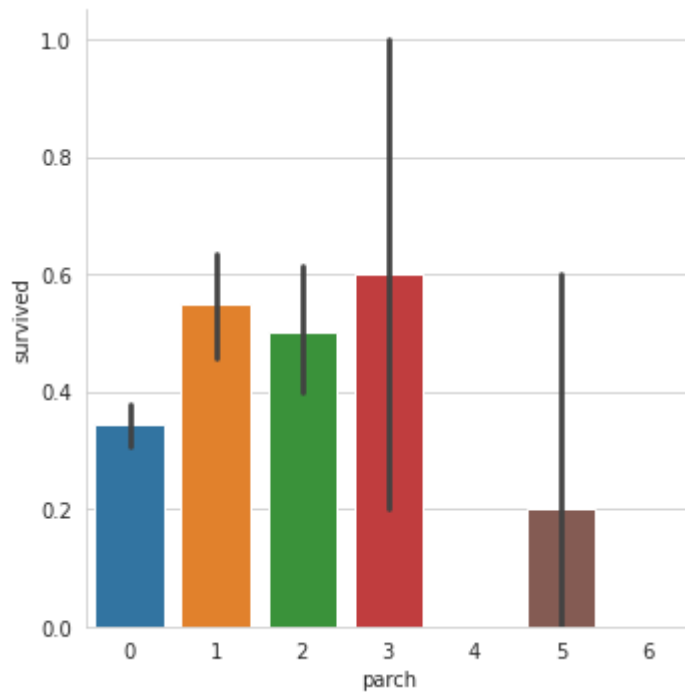
```
sns.catplot(x='survived', y='embark_town',  
            hue='class', kind='bar',  
            data=titanic);
```



```
sns.catplot(x='sibsp', y='survived',  
            kind='bar', data=titanic);
```



```
sns.catplot(x='parch', y='survived',  
            kind='bar', data=titanic);
```



```
sns.catplot(x='alone', y='survived',  
            kind='bar', data=titanic);
```

